

RESEARCH



Viewpoint-invariant exercise repetition counting

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Abstract

Counting the repetition of human exercise and physical rehabilitation is common in rehabilitation and exercise training. The existing vision-based repetition counting methods less emphasize the concurrent motions in the same video, and counting skeleton in different view angles. This work analyzed the spectrogram of the pose estimation cosine similarity to count the repetition. Besides the public datasets. This work also collected exercise videos from 11 adults to verify that the proposed method can handle concurrent motion and different view angles. The presented method was validated on the University of Idaho Physical Rehabilitation Movements Data Set (UI-PRMD) and MM-fit dataset. The overall mean absolute error (MAE) for MM-fit was 0.06 with off-by-one Accuracy (OBOA) of 0.94. As for the UI-PRMD dataset, MAE was 0.06 with OBOA 0.95. We have also tested the performance in various camera locations and concurrent motions with 57 skeleton time-series videos with an overall MAE of 0.07 and OBOA of 0.91. The proposed method provides a view-angle and motion agnostic concurrent motion counting. This method can potentially use in large-scale remote rehabilitation and exercise training with only one camera.

Keywords: Camera, Repetition counting, Exercise

Introduction

Repetitive exercise and training are omnipresent in daily life and are especially common in sport and rehabilitation training. Early research [1] has surveyed that disobedience to the exercise program is a factor that deteriorate the training outcome. Some researchers [2, 3] believe that lack of motivation and feedback is why the users do not comply with the designated exercise plan. Engaging users by providing more feedback and motivating results could increase the adherence of the users [4, 5], and consequently, enhance the training outcome.

As technology advances, training and rehabilitation programs might not necessarily occur in primary care. The concept of the home exercise program (HEP) and remote rehabilitation with information technology assistance draws researchers' and medical professionals' attention. This approach relieves the medical area's

burden and enables flexibility for people who encounter difficulty getting medical resources. Providing repetition counting feedback is common feedback in the HEP. Previous works [6–9] focused on HEP systems also implemented several feedbacks, such as repetition counting. Users can be more engaged in the exercise and training program given the counting feedback. Users could know how many repetitions have been done and their speed for completing a cycle.

Still, counting exercise repetition is a less studied than the prosperous-growing field in human activity recognition. Dwibedi et al. [10] pointed out that searching for suitable video from large-scale public datasets is strenuous. No specific keyword and annotations cater to this particular need. The root cause of this phenomenon is that annotating video or signal across the temporal domain is labor-intensive and monotonous work. Due to these reasons, the repetition counting task usually encounters the lack of sufficient annotated data.

This research aims to solve a counting on concurrent human exercise motions. This research provides a novel repetition counting method using skeleton data based on any camera, including the 3D depth camera. The

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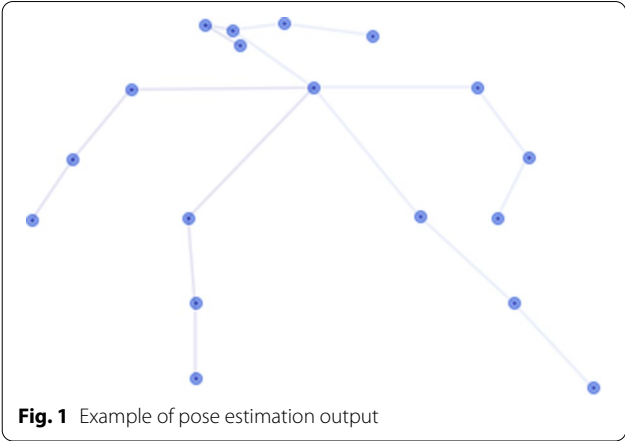
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proposed method first calculates the self-similarity of the sequence of skeleton data generated from pose estimation methods. Subsequently, We analyzed the spectrogram of the self-similarity and estimated the count.

The primary contribution of this work is that it offers (1) view-angle and motion agnostic vision-based counting methods. This work takes the skeleton data from the camera and counts through the temporal similarity. (2) Counting multiple people concurrent repetition in the same frame. The proposed method takes the skeleton data as the input. Current pose estimation methods can estimate multiple people in a frame, so the proposed method could leverage these advantages and infer the repetition for each person with different repetition frequencies in the same video. The current state of the art [10, 11] usually has the limitation of counting only one motion in the videos. In contrast, counting concurrent movement becomes possible by counting skeleton data. This work has strong application in group exercise training and human factors work-study.

The current research works can be categorized using these two aspects, the feature used for counting and counting methods, listed in Table 1. In terms of features used for counting, the current study can be categorized into 1. Crafted features, 2. Self-similarity features. The other aspect is the counting method being implemented. Current counting methods can be categorized into three approaches 1. Peak detection, 2. Frequency transformation and 3. Neural network methods. Recent research works can be classified as a combination of these two aspects (Fig. 1).

In the feature perspectives, only pose estimation features can identify several different human movements in the video and can estimate multiple periodical movements in the videos. Current pose estimation [17–22] will output a connected graph of the human joints. Figure An illustration of the pose estimation output. Previous studies [13, 14] leverage these ideas to count exercise repetitions, but this work uses handcrafted features for different exercises. This disadvantage makes this approach hard to generalize all periodical movements. Self-similarity features for counting were only used in previous



research [10]. However, the self-similarity features are popular and proved to be view-angle agnostic in human motion recognition fields [23–25]. Dwibedi et al. [10] study cannot count people exercising in the same video together as it does not consider such a situation.

From the counting method perspective, most methods adopt either frequency method or peak counting methods. Peak detection methods are intuitive but subject to the noise in the signal. Frequency domain counting methods are more robust to the signal noise and with fewer tuning parameters. In addition, Dwibedi et al. [10] work using neural network methods from the self-similarity features to count the repetition has drawn tremendous success and opened another approach to solve this challenge.

The proposed algorithm is a self-similarity feature with a frequency-based counting method. From the previous literature, we can observe that this method has the advantages of fewer tuning parameters and view-angle agnostics.

Method

Data source

Our method was trained and tested using two public datasets, the UI-PRMD dataset [26] and MM-fit [14]. These datasets give each video a label indicating how many repetitions are in the video. In addition to these two datasets, an additional dataset was collected to verify the claim of viewpoint-invariant and concurrent motion.

UI-PRMD dataset

UI-PRMD dataset [26] is composed of ten motions commonly used in rehabilitation, including (1) deep squat, (2) hurdle step, (3) inline lunge, (4) side lunge, (5) sit to stand, (6) standing active straight leg raise, (7) standing shoulder abduction, (8) standing shoulder extension, (9) standing shoulder internal-external rotation, and

Table 1 Counting method used in the previous research, method using pose estimation is in bold

	Crafted features	Self-similarity features
Peak detection	[12–14]	NA
Frequency domain counting	[15, 16]	This study
Neural network	NA	[10]

(10) standing shoulder scaption. Ten healthy subjects repeated each exercise 10 times. A Kinect v2 camera was put in front of the subject during the data collection. It is worth pointing out that subjects were allowed to use their dominant side to perform the tasks, so some might use the right side, and some may use the left side to complete tasks 2, 3, 4, 6, 7, 8, 9, and 10.

MM-fit dataset

The MM-fit dataset [14] consists of ten types of commonly used workouts for home training (1) squats, (2) push-ups, (3) shoulder press, (4) lunges, (5) dumbbell rows, (6) sit-ups, (7) triceps extensions, (8) biceps curls, (9) lateral raises, (10) and jumping jacks. There are ten participants in the dataset, and each participant was asked to perform a set of exercises consisting of several exercises with ten repetitions. An RGB depth camera was put in front of the participant during the training. This dataset adopted OpenPose for 2D pose estimation and the method developed by Martina et al. [20] for 3D pose estimation.

Additional dataset

We acknowledged that both public datasets fixed the camera location and did not test concurrent motions in the same videos. 11 adults were recruited and asked to squat, lateral shoulder raises, and lunge for ten repetitions with different camera angles or concurrently to demonstrate the proposed idea work in the various scenarios. Total 57 skeleton time series were extracted from the videos, and the detailed decomposition was listed in Table 2. Videos were recorded by iPad Pro 5th generation with a frame rate of 30 Hz, and the pose estimation was done by OpenPose [21]. Please refer to the supplementary animated GIF picture for a typical example of identifying the motions of two people. The counting label for each label was labeled by two student annotators, and the kappa agreement is 1.00 between two student annotators. This research was approved by the Human Subjects Ethics Sub-Committee, City University of Hong Kong (Ref. 3-2-201803_02).

Outline of the proposed method

The proposed method relied on the existing pose estimation algorithms to extract the skeleton location of each frame. Though the proposed method relied on the pose estimation, it did not rely on a specific type of skeleton format. Consequently, 3D pose skeleton formats like Kinect or 2D skeleton in Openpose are all valid input for this method.

After obtaining skeleton data, this work proposed a new perspective of processing the skeleton graph time series to construct the temporal self-similarity.

Table 2 Description of the additional data

Description	Number of videos
Camera location	
Front	20
Left	19
Right	18
Concurrent/single motion	
Single person	36
More than one	21
Exercise	
Squat	25
Lateral shoulder raise	16
Lunge	16

Subsequently, we constructed a spectrogram for the pairwise similarity and counted the repetition by integrating it from the spectrogram.

Pose estimation

UI-PRMD dataset [26] used Microsoft Kinect v2 built-in 3D pose estimation, which estimated 22 joint locations in the 3D distance from the camera. MM-fit [14] dataset provides 2D pose estimation and 3D pose estimation by Openpose or Martina et al. [20]. Neither MM-fit nor UI-PRMD dataset provides original video, but it is worth emphasizing that the proposed method is vision-based.

Cosine similarity calculation

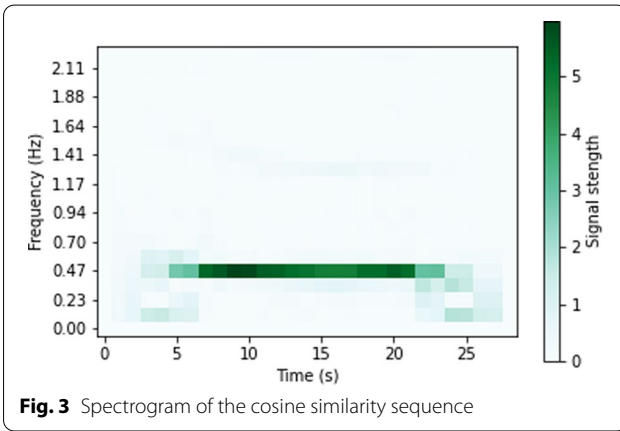
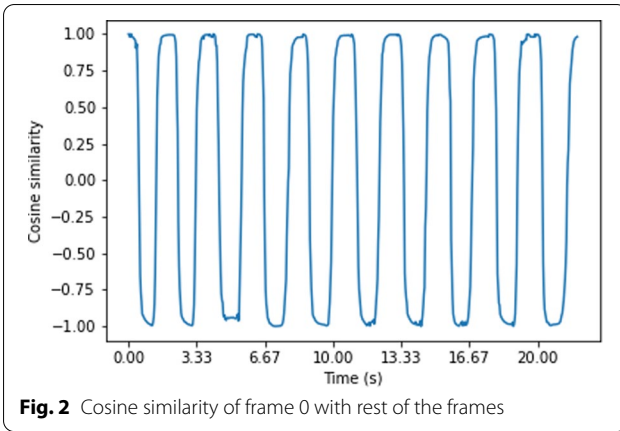
After getting the skeleton location, we first compute the pairwise cosine similarity of the skeleton data. We assume most of the training exercises are back-and-forth movements starting from the frame 0. These back-and-forth movements can be conceptualized through cosine similarity to quantify it as if performing the simple harmonic motion. This assumption is valid for movements in public datasets. Given a skeleton data time series X with length t , X was a matrix with size $(t, j \times d)$, where j was the amount of joint, and d was either 2 or 3 representing the dimension of the data. The self cosine similarity is the normalized dot product between the observation at time t_0 and time t_i .

$$\text{self cosine similarity} = \cos(\theta) = \frac{X_{t_0} \cdot X_{t_i}}{\|X_{t_0}\| \|X_{t_i}\|} \quad (1)$$

Figure 2 displays the output similarity corresponding to time 0.

Constructing spectrogram

Observing the strong periodicity, analyzing the frequency patterns, and counting through the frequency



domain patterns should be reasonable. We started to investigate the spectrogram of cosine similarity for $t = 0$ with respect to the rest of the time point i , $\forall i \leq t$. The sequence is demonstrated in Fig. 2. The spectrogram was done by calculating the fast Fourier transform (FFT) on a fixed sliding window with window size w . For signals x in the sliding window segment, the FFT response:

$$\text{FFT}(x)_k = \sum_{n=0}^{w-1} x(n) \exp\left(\frac{-i2\pi n}{w}\right), k = 0, \dots, w-1 \quad (2)$$

Figure 3 illustrates the spectrogram of the self-cosine similarity in Fig. 2. The yellow line in Fig. 3 revealed a strong signal at the frequency of around 0.25 along with the whole exercise.

Counting out of the time

We believe the most dominant frequency in the spectrogram is the frequency corresponding to the repeating motion. The local frequency f_m in the sliding window

m is the frequency with the highest amplitude in such a sliding window.

$$f_m = \arg\max_k \|\text{FFT}(x)_k\| \quad (3)$$

The estimation of the repetition counts $\hat{c}$ was defined as the integral of local frequency over time, where δt is the time length in that FFT window.

$$\hat{c} = \sum_{t=t_0}^{t_i} f_t \delta t \quad (4)$$

Runia et al. [15] adopted the same counting method, which emphasized counting non-stationary signals.

Consideration of hyperparameters

The only hyperparameter pair in this method was the FFT spectrogram's sliding window size and sliding steps. There are two factors to take into consideration. First, the sliding window should be wide enough to envelop at least a cycle of repetition. Therefore, the lower bound of the window size should be the frame rate times the maximum length of a cycle. This lower bound is at 90 in the public dataset used in this study. Besides, the wider the window, the better the frequency resolution. As a trade-off, wider windows and more giant sliding window steps resulted in lower time resolution. Both public datasets' skeleton sequences are around 400 to 800 frames, so we choose the sliding window size 128, 256 for testing. We have evaluated and discussed the different combinations of sliding window sizes and steps in the result and discussion.

Model evaluation

There are two evaluation metrics commonly used [10, 11, 15] in repetition counting.

Mean absolute error (MAE) of the count

MAE of the count is the average normalized absolute difference between the predicted count on the i -th video $\hat{c}_i$ and the ground truth c_i of the i -th video. This metric was used in the previous research works as the metric can be interpreted as the percentage of the counting difference compared with the ground truth.

$$\text{MAE of the count} = \frac{1}{n} \sum_{i=1}^n \frac{|\hat{c}_i - c_i|}{c_i}$$

Off-by-one accuracy (OBOA)

OBOA first labels the video is correctly classified if the absolute difference between the predicted count and the ground truth is less or equal to one. Otherwise, it marks the prediction as misclassified. The OBOA is the

accuracy of using such a definition. OBOA provides a brief idea of how accurate the algorithm is and favors a precise algorithm with a tolerance of a few extremely miscounting cases.

$$\text{OBOA} = \frac{1}{n} \sum_{i=1}^n n_i, \quad n_i = \begin{cases} 1 & \text{if } |\hat{c}_i - c_i| \leq 1 \\ 0 & \text{else} \end{cases}$$

Results

Overall, the proposed method can reach 0.94 to 0.95 OBOA with MAE 0.06 with the best hyperparameter combinations.

MM-fit dataset

Table 3 indicated the MAE and OBOA for the proposed method and the baseline demonstrated in [14]. The MAE and OBO error was reported with the same testing scheme as Stromback et al. [14]. The proposed method yielded the same or significantly better in each task. In addition to the baseline method, the proposed method performs more stably across different types of motion. In contrast, the baseline method varies significantly among different kinds of exercise. This might be due to the baseline method requiring adjusting the hyperparameter specific to the motion.

The MM-fit dataset also provided 2D pose estimation data using Openpose [21]. We have listed the performance of the MM-fit dataset using 2D pose estimation format as the input in Table 4.

UI-PRMD dataset

Table 5 indicates the MAE and OBOA for the proposed method testing on the UI-PRMD dataset. All the data were used to report the result. To the best of our knowledge, there is no research work adopted this dataset for motion counting.

Additional dataset

Table 6 was obtained using 256 window sizes with step 1. Table 6 indicated that the proposed method was able to count with different view angles and concurrent motions. The boxplots (Fig. 4) of the MAE suggest that at least 75% of the pose estimation time series were miscounted within one repetition for any situation. Only a few video clips showed extremely incorrect estimations.

Discussion

Effect of the hyperparameters

Tables 3 and 5 have demonstrated the effect of hyperparameters. We could observe that a window size of 256 was more desirable than 128 (MANOVA, $p < 0.001$). Window size with length 256 performed better than window size with 128 in overall MAE and OBOA. The only exception was jumping jacks in the MM-fit dataset. We suspected the primary reason was that performing jumping jacks took significantly less time than the rest of the exercise (student t-test, $p < 0.001$). The average frame length of jumping jacks was 309 frames with a standard deviation of 51, and in contrast, the rest of the exercise took 637 frames with a standard deviation of 163. Consequently, a window with 256 frames might not provide

Table 3 MAE of count and OBOA for motion counting MM-fit dataset, squats (Sq), push-ups (Pu), dumbbell shoulder press (Sp), lunges (Lg), dumbbell rows (Dr), situps (Su), tricep extensions (Te), bicep curls (Bc), lateral shoulder raises (Sr), and jumping jacks (Jj)

Windows steps	128					256					Baseline
	1	2	4	8	16	1	2	4	8	16	
MAE for each motion											
Sq	0.10	0.10	0.10	0.10	0.10	0.05	0.05	0.05	0.05	0.06	0.05
Pu	0.08	0.08	0.08	0.07	0.07	0.09	0.08	0.09	0.09	0.10	0.52
Sp	0.08	0.08	0.08	0.08	0.06	0.03	0.03	0.03	0.04	0.04	0.63
Lg	0.10	0.10	0.10	0.10	0.10	0.04	0.04	0.04	0.04	0.03	0.03
Dr	0.06	0.06	0.06	0.06	0.07	0.06	0.06	0.06	0.06	0.06	0.23
Su	0.11	0.11	0.12	0.12	0.12	0.11	0.11	0.11	0.11	0.11	0.17
Te	0.07	0.07	0.07	0.07	0.07	0.05	0.06	0.06	0.06	0.06	0.33
Bc	0.07	0.07	0.07	0.06	0.07	0.05	0.05	0.05	0.05	0.05	0.41
Sr	0.08	0.08	0.08	0.08	0.08	0.04	0.04	0.04	0.04	0.04	0.09
Jj	0.03	0.03	0.04	0.05	0.06	0.07	0.07	0.07	0.07	0.09	0.26
Overall											
MAE	0.08	0.08	0.08	0.08	0.08	0.06	0.06	0.06	0.06	0.06	0.67
OBOA	0.89	0.89	0.89	0.89	0.89	0.94	0.94	0.93	0.92	0.91	0.90

Table 4 MAE of count for motion counting MM-fit dataset (2D), squats (Sq), push-ups (Pu), dumbbell shoulder press (Sp), lunges (Lg), dumbbell rows (Dr), situps (Su), tricep extensions (Te), bicep curls (Bc), lateral shoulder raises (Sr), and jumping jacks (Jj)

Windows steps	128					256				
	1	2	4	8	16	1	2	4	8	16
MAE for each motion										
Sq	0.11	0.11	0.10	0.11	0.11	0.06	0.06	0.06	0.05	0.07
Pu	0.10	0.10	0.10	0.09	0.09	0.13	0.14	0.13	0.14	0.14
Sp	0.10	0.10	0.09	0.09	0.07	0.04	0.04	0.04	0.05	0.05
Lg	0.10	0.10	0.10	0.10	0.10	0.04	0.04	0.04	0.04	0.04
Dr	0.07	0.07	0.06	0.06	0.07	0.06	0.06	0.05	0.04	0.05
Su	0.22	0.22	0.21	0.21	0.21	0.17	0.17	0.17	0.16	0.17
Te	0.07	0.07	0.08	0.08	0.08	0.06	0.06	0.06	0.06	0.06
Bc	0.07	0.07	0.07	0.07	0.07	0.05	0.05	0.05	0.05	0.05
Sr	0.08	0.08	0.08	0.08	0.08	0.04	0.04	0.04	0.04	0.03
Jj	0.04	0.04	0.04	0.05	0.06	0.13	0.14	0.14	0.14	0.15
Overall										
MAE	0.09	0.09	0.09	0.09	0.09	0.08	0.08	0.08	0.08	0.08
OBOA	0.87	0.87	0.88	0.88	0.87	0.87	0.87	0.87	0.86	0.85

Table 5 MAE of count and OBOA for motion counting UI-PRMD dataset, deep squat (Dq), hurdle step (Hs), inline lunge (Il), side lunge (Sl), sit to stand (Ss), leg raise (Lr), shoulder abduction (Sa), shoulder extension (Se), shoulder internal-external rotation (Sr), and shoulder scaption (Sc)

Windows steps	128					256				
	1	2	4	8	16	1	2	4	8	16
MAE for each motion										
Dq	0.07	0.07	0.08	0.09	0.08	0.05	0.05	0.05	0.04	0.04
Hs	0.10	0.10	0.09	0.09	0.07	0.00	0.01	0.01	0.01	0.02
Il	0.10	0.10	0.10	0.10	0.09	0.04	0.04	0.04	0.04	0.05
Sl	0.14	0.14	0.14	0.12	0.12	0.07	0.07	0.07	0.07	0.05
Ss	0.12	0.12	0.11	0.11	0.09	0.02	0.02	0.02	0.04	0.05
Lr	0.15	0.16	0.15	0.15	0.14	0.11	0.11	0.10	0.10	0.11
Sa	0.08	0.08	0.09	0.09	0.09	0.05	0.05	0.05	0.06	0.05
Se	0.11	0.11	0.11	0.11	0.12	0.12	0.12	0.12	0.12	0.13
Sr	0.08	0.08	0.09	0.09	0.09	0.08	0.08	0.08	0.07	0.05
Sc	0.05	0.05	0.04	0.04	0.03	0.06	0.06	0.06	0.06	0.06
Overall										
MAE	0.10	0.10	0.10	0.10	0.09	0.06	0.06	0.06	0.06	0.06
OBOA	0.81	0.80	0.82	0.82	0.87	0.94	0.94	0.94	0.94	0.94

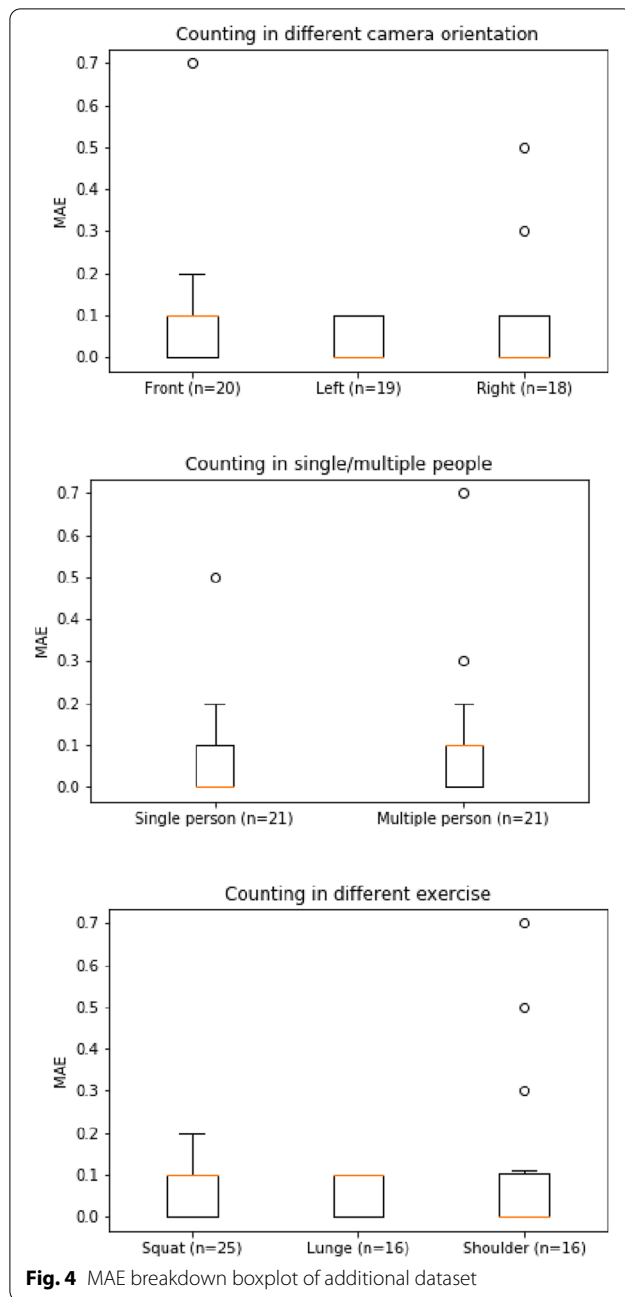
sufficient time resolution to estimate the frequency accurately. For the rest of the exercise, 256 window size offered a higher frequency resolution than 128 windows to count the repetition more accurately than 128. For the same reason, a sliding window with size 512 is not suitable for all of the exercises in the public datasets.

Sliding window step size does not affected the repetition counting result (MANOVA, $p = 0.305$). Table 3 showed that MAE increased and OBOA decreased while

increasing the step size in the MM-fit dataset. Interestingly, this effect cannot be observed in the UI-PRMD dataset (Table 5). In any case, the result shown in Tables 3 and 5 indicated that sliding window steps made a minor difference in repetition counting.

Robustness of the proposed method

The robustness of our proposed method has been tested. The robustness is vital for the skeleton-based method as



pose estimation performance is subject to occlusion from the environments and the view-angle. Consequently, noisy pose estimation or missing frames were expected in the real-world application: two scenarios, 1. A zero-mean Gaussian noise with different standard deviation levels and 2. missing data were added to the public datasets to test the robustness of the proposed method. The standard deviation of the Gaussian noise was set on different percentages of the corresponding axis and joint. The missing data was simulated by replacing the frames

Table 6 Counting result of the additional data

Description	MAE (SD)	OBOA
Camera location		
Front	0.11 (0.15)	0.85
Left	0.04 (0.05)	1.00
Right	0.078 (0.13)	0.89
Concurrent/single motion		
Single person	0.05 (0.09)	0.94
More than one	0.11 (0.15)	0.85
Exercise		
Squat	0.06 (0.06)	0.92
Lateral shoulder raise	0.11 (0.20)	0.81
Lunge	0.06 (0.05)	1.00

with interpolation from the nearest frames. Figures 5 and 6 showed how these scenarios affected the proposed method.

Figure 5 shows the performance of the proposed method while Gaussian noises were present. The Gaussian noise follows a zero-mean Gaussian distribution with the standard deviation 0% to 100% of the standard deviation of the corresponding joint axis. The MAE increased while the noise increased. Figure 5 revealed that a wider sliding window (window size = 256) is more robust to the Gaussian noise. The MAE did not increase significantly within increasing 30% of standard deviation 256 sliding window size (MANOVA, $p = 0.97$).

Our method is barely affected by the missing data. Figure 6 shows the performance of the proposed method while some parts of the frames are missing. This scenario could be treated as the pose estimation method failing to recognize the person while there was a person in the frame. In each simulation, a percentage of frames were replaced by a zero vector to simulate the missing. Subsequently, these missing frames were imputed by interpolating the nearest frame before the loss and after the loss. Under this imputation method, the proposed method remained nearly constant MAE (MANOVA, $p > 0.99$). The primary reason is that interpolating the missing data did not significantly alter the periodical patterns of the skeleton, so the performance of the proposed method did not vary while missing data was present.

Counting in different types of skeleton formats

We have also illustrated counting in different types of skeleton formats. MM-fit data is in either a 3D COCO skeleton format (18 joints, 3D) or a 2D COCO skeleton format (18 joints, 2D). UI-PRMD dataset uses Microsoft Kinect skeleton format, which has 22 3D joint locations. The data collected in this study was in 2D COCO

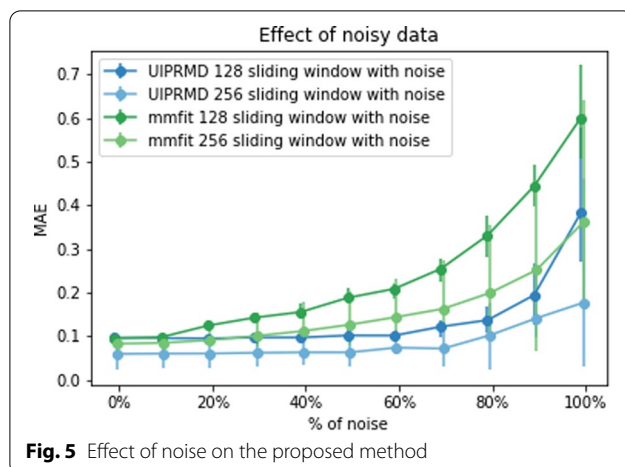


Fig. 5 Effect of noise on the proposed method

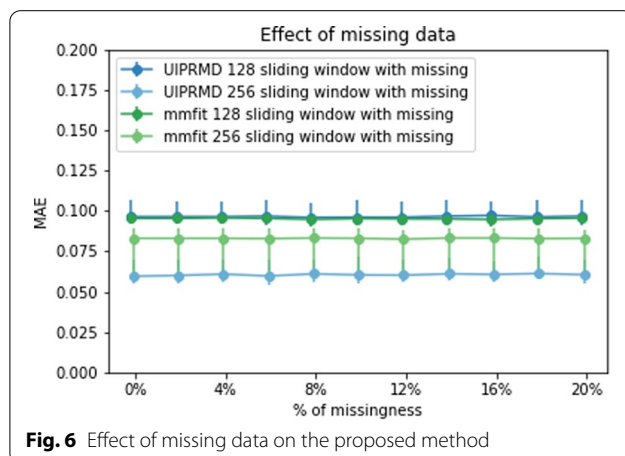


Fig. 6 Effect of missing data on the proposed method

skeleton format. The proposed method all exhibited satisfactory results. Table 4 indicated that counting based on the 2D skeleton format would deteriorate the OBOA and MAE compared with 3D pose estimation inputs. We could observe that counting in the 2D will deteriorate the result compared with using 3D. This deterioration is pronounced in push-ups, situps, and jumping jacks. The reason for this variance among different exercises was not that clear, and further investigation might be helpful.

Potential application

This work could potentially be used in the exercise and rehabilitation training, especially in the group session. The proposed method accurately counts repetitive motion to measure movements from several people in a single camera. This advantage could facilitate monitoring exercise outcomes in the group class or open area. In addition to healthcare application, this method could also be adopted in work design to effectively analyze human performance in repetitive tasks.

The novelty of the proposed work

This work provides a new perspective on analyzing human repetition counting. There is a new finding and a breakthrough in this research. There is no research investigating the frequency domain pattern of pose estimation results to the best of the author's knowledge. We verify that the repetitive exercise which shows periodical patterns in the video could be analyzed through Fourier transform. This research tackles counting different periodical exercises in the same video through the finding. This task is less addressed in similar research.

Limitation and future research direction

This method assumes the motion is a simple back and forth exercise in which the similarity of skeleton data behaves like periodical signals. Exercise with more complicated procedures might not demonstrate this property. Nonetheless, most rehab or exercise training does not consist of this kind of complex exercise. Second, we assume the most dominant frequency corresponds to the exercise. Minor motion counting will need further investigation on the spectrogram. The last limitation is that we rely on motion detection and recognition repetition motion to perform the proposed algorithm. We might investigate more exercises in different contexts to understand how these limitations affect the proposed method.

Conclusion

We have presented a motion repetition counting method using skeleton data. This research provides a solution to concurrent repetition counting in human exercise, which is less addressed in the previous study. The proposed method has been verified on the public datasets and a few conveniently collected videos with desirable accuracy in different view angles, motions, and concurrent motions. The proposed method possesses the potential in rehabilitation and exercise training to monitor progress and provide remote rehabilitation and exercise training.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s13755-023-00258-3>.

Below is the link to the electronic supplementary material. Supplementary file1 (GIF 5363 kb).

Author contributions

YCH conducted the analysis and writing of the report and data collection. YCH, TE, and KT contributed to the study design and review of the manuscript.

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Data availability

The data supporting this study's findings are available on request from the corresponding author, YCH. The data are not publicly available due to the privacy of research participants.

Code availability

The code for this work is available on <https://github.com/YuChengHSU/repetition-counting>.

Declarations

Conflict of interest

The authors declare that they have no competing interests.

Ethical approval

This research was approved by the Human Subjects Ethics Sub-Committee, City University of Hong Kong (Ref. 3-2-201803_02). All of the participants were well-informed and consent to participate the experiment.

Consent for publication

Written informed consent for publication was obtained from all of the participants in our experiment.

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